Table 1: The governing discrete equations with sample time h.

continuity:

$$0 = -\rho(\delta y_{j,j+1}\,(u_{i+1,J} - u_{i,J}) + \delta x_{i,i+1}\,(v_{I,j+1} - v_{I,j}))$$
$$\forall i = I \in \{2, N_x - 1\}\ \ \forall j = J \in \{2, N_y - 1\}$$

x-momentum:

$$a^{px}_{i,J} u_{i,J} = a^{nx}_{i,J} u_{i,J+1} + a^{sx}_{i,J} u_{i,J-1} + a^{wx}_{i,J} u_{i-1,J} + a^{ex}_{i,J} u_{i+1,J} - (p_{I,J} - p_{I-1,J})\,\delta y_{j,j+1} + S^x_{i,J} + c_x u^0_{i,J}$$
$$\forall i = I \in \{3, N_x - 1\}\ \ \forall j = J \in \{2, N_y - 1\}$$

y-momentum:

$$a^{py}_{I,j} v_{I,j} = a^{ny}_{I,j} v_{I,j+1} + a^{sy}_{I,j} v_{I,j-1} + a^{wy}_{I,j} v_{I-1,j} + a^{ey}_{I,j} v_{I+1,j} - (p_{I,J} - p_{I,J-1})\,\delta x_{i,i+1} + S^y_{I,j} + c_y v^0_{I,j}$$
$$\forall i = I \in \{2, N_x - 1\}\ \ \forall j = J \in \{3, N_y - 1\}$$

where

$$a^{wx}_{i,J} = \max\left[F^{wx}_{i,J}, \left(D^{wx}_{i,J} + \tfrac{F^{wx}_{i,J}}{2}\right), 0\right] \qquad a^{sx}_{i,J} = \max\left[F^{sx}_{i,J}, \left(D^{sx}_{i,J} + \tfrac{F^{sx}_{i,J}}{2}\right), 0\right] + T^{sx}_{i,J}$$
$$a^{ex}_{i,J} = \max\left[-F^{ex}_{i,J}, \left(D^{ex}_{i,J} - \tfrac{F^{ex}_{i,J}}{2}\right), 0\right] \qquad a^{nx}_{i,J} = \max\left[-F^{nx}_{i,J}, \left(D^{nx}_{i,J} - \tfrac{F^{nx}_{i,J}}{2}\right), 0\right] + T^{nx}_{i,J}$$

$$a^{px}_{i,J} = a^{nx}_{i,J} + a^{ex}_{i,J} + a^{sx}_{i,J} + a^{wx}_{i,J} + F^{nx}_{i,J} + F^{ex}_{i,J} - F^{sx}_{i,J} - F^{wx}_{i,J} + T^{px}_{i,J} + a^{px}_0$$

$$a^{wy}_{I,j} = \max\left[F^{wy}_{I,j}, \left(D^{wy}_{I,j} + \tfrac{F^{wy}_{I,j}}{2}\right), 0\right] \qquad a^{sy}_{I,j} = \max\left[F^{sy}_{I,j}, \left(D^{sy}_{I,j} + \tfrac{F^{sy}_{I,j}}{2}\right), 0\right]$$
$$a^{ey}_{I,j} = \max\left[-F^{ey}_{I,j}, \left(D^{ey}_{I,j} - \tfrac{F^{ey}_{I,j}}{2}\right), 0\right] \qquad a^{ny}_{I,j} = \max\left[-F^{ny}_{I,j}, \left(D^{ny}_{I,j} - \tfrac{F^{ny}_{I,j}}{2}\right), 0\right]$$

$$a^{py}_{I,j} = a^{ny}_{I,j} + a^{ey}_{I,j} + a^{sy}_{I,j} + a^{wy}_{I,j} + F^{ny}_{I,j} + F^{ey}_{I,j} - F^{sy}_{I,j} - F^{wy}_{I,j} + a^{py}_0$$

in which:

$$F^{wx}_{i,J} = \tfrac{1}{2}\rho\,(u_{i,J} + u_{i-1,J})\,\delta y_{j,j+1} \qquad D^{wx}_{i,J} = \mu \tfrac{\delta y_{j,j+1}}{\Delta x_{I-1,I}} \qquad c_x = a^{px}_0 = \tfrac{\Delta x_{I-1,I}\delta y_{j,j+1}}{h}$$
$$F^{ex}_{i,J} = \tfrac{1}{2}\rho\,(u_{i+1,J} + u_{i,J})\,\delta y_{j,j+1} \qquad D^{ex}_{i,J} = \mu \tfrac{\delta y_{j,j+1}}{\Delta x_{I-1,I}} \qquad \Delta x_{I-1,I} = x_I - x_{I-1}$$
$$F^{sx}_{i,J} = \tfrac{1}{2}\rho\,(v_{I,j} + v_{I-1,j})\,\Delta x_{I-1,I} \qquad D^{sx}_{i,J} = \mu \tfrac{\Delta x_{I-1,I}}{\delta y_{j,j+1}} \qquad \delta y_{j,j+1} = y_{j+1} - y_j$$
$$F^{nx}_{i,J} = \tfrac{1}{2}\rho\,(v_{I,j+1} + v_{I-1,j+1})\,\Delta x_{I-1,I} \qquad D^{nx}_{i,J} = \mu \tfrac{\Delta x_{I-1,I}}{\delta y_{j,j+1}}$$
$$F^{wy}_{I,j} = \tfrac{1}{2}\rho\,(u_{i,J} + u_{i,J-1})\,\Delta y_{J-1,J} \qquad D^{wy}_{I,j} = \mu \tfrac{\Delta y_{J-1,J}}{\delta x_{i,i+1}} \qquad c_y = a^{py}_0 = \tfrac{\Delta y_{J-1,J}\delta x_{i,i+1}}{h}$$
$$F^{ey}_{I,j} = \tfrac{1}{2}\rho\,(u_{i+1,J} + u_{i+1,J-1})\,\Delta y_{J-1,J} \qquad D^{ey}_{I,j} = \mu \tfrac{\Delta y_{J-1,J}}{\delta x_{i,i+1}} \qquad \Delta y_{J-1,J} = y_J - y_{J-1}$$
$$F^{sy}_{I,j} = \tfrac{1}{2}\rho\,(v_{I,j-1} + v_{I,j})\,\delta x_{i,i+1} \qquad D^{sy}_{I,j} = \mu \tfrac{\delta x_{i,i+1}}{\Delta y_{J-1,J}} \qquad \delta x_{i,i+1} = x_{i+1} - x_i$$
$$F^{ny}_{I,j} = \tfrac{1}{2}\rho\,(v_{I,j} + v_{I,j+1})\,\delta x_{i,i+1} \qquad D^{ny}_{I,j} = \mu \tfrac{\delta x_{i,i+1}}{\Delta y_{J-1,J}}$$

$$T^{nx}_{i,J} = \tfrac{\rho l_m^2 \Delta x_{I-1,I}}{(\Delta y_{J,J+1})^2}\,|u_{i,J+1} - u_{i,J}|, \qquad T^{sx}_{i,J} = \tfrac{\rho l_m^2 \Delta x_{I-1,I}}{(\Delta y_{J-1,J})^2}\,|u_{i,J} - u_{i,J-1}|, \qquad T^{px}_{i,J} = T^{sx}_{i,J} + T^{nx}_{i,J}$$

$a^{wx}(u_{i-1,J}, u_{i,J})$, $a^{ex}(u_{i,J}, u_{i+1,J})$, $a^{nx}(v_{I-1,j+1}, v_{I,j+1}, u_{i,J}, u_{i,J+1})$, $a^{sx}(v_{I-1,j}, v_{I,j}, u_{i,J-1}, u_{i,J})$,
$a^{px}(u_{i-1,J}, u_{i,J-1}, u_{i,J}, u_{i,J+1}, u_{i+1,J}, v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1})$,
$a^{wy}(u_{i,J-1}, u_{i,J})$, $a^{ey}(u_{i+1,J-1}, u_{i+1,J})$, $a^{ny}(v_{I,j+1}, v_{I,j})$, $a^{sy}(v_{I,j}, v_{I,j-1})$,
$a^{py}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1})$

The equations in Table 1 can be reformulated in matrix form by taking the following vectors:

$$u_k = \begin{bmatrix} u_{3,2} \\ \vdots \\ u_{3,N_y-1} \\ u_{4,2} \\ \vdots \\ u_{4,N_y-1} \\ \vdots \\ u_{N_x-1,2} \\ \vdots \\ u_{N_x-1,N_y-1} \end{bmatrix}, \; v_k = \begin{bmatrix} v_{2,3} \\ \vdots \\ v_{2,N_y-1} \\ v_{3,3} \\ \vdots \\ v_{3,N_y-1} \\ \vdots \\ v_{N_x-1,3} \\ \vdots \\ v_{N_x-1,N_y-1} \end{bmatrix}, \; p_k = \begin{bmatrix} p_{2,2} \\ \vdots \\ p_{2,N_y-1} \\ p_{3,2} \\ \vdots \\ p_{3,N_y-1} \\ \vdots \\ p_{N_x-1,2} \\ \vdots \\ p_{N_x-1,N_y-1} \end{bmatrix},$$

where $u_k \in \mathbb{R}^{(N_x-3)(N_y-2)}$, $v_k \in \mathbb{R}^{(N_x-2)(N_y-3)}$ and $p_k \in \mathbb{R}^{(N_x-2)(N_y-2)}$ are the vectors that stack all the velocities and pressures in every point of the staggered grid. Now we can obtain the following set of matrix equations which we need to solve:

$$\begin{bmatrix} A_x(u_k, v_k) & 0 & B_1 \\ 0 & A_y(u_k, v_k) & B_2 \\ B_1^T & B_2^T & 0 \end{bmatrix} \begin{bmatrix} u_{k+1} \\ v_{k+1} \\ p_{k+1} \end{bmatrix} = \begin{bmatrix} b_x(u_b, u_k) + c_x u_k + S^x(u_k, v_k) \\ b_y(v_b, u_k) + c_y v_k + S^y(u_k, v_k) \\ b_c(u_b, v_b) \end{bmatrix} \tag{1}$$

$$\tag{2}$$

with $k = 0, 1, \ldots, N$ representing discrete time. The boundary velocities are u_b and v_b. Note that (for example) $i = I$ begins at 3 in the x-momentum equations while it begins at 2 in the y-momentum equation.

Example

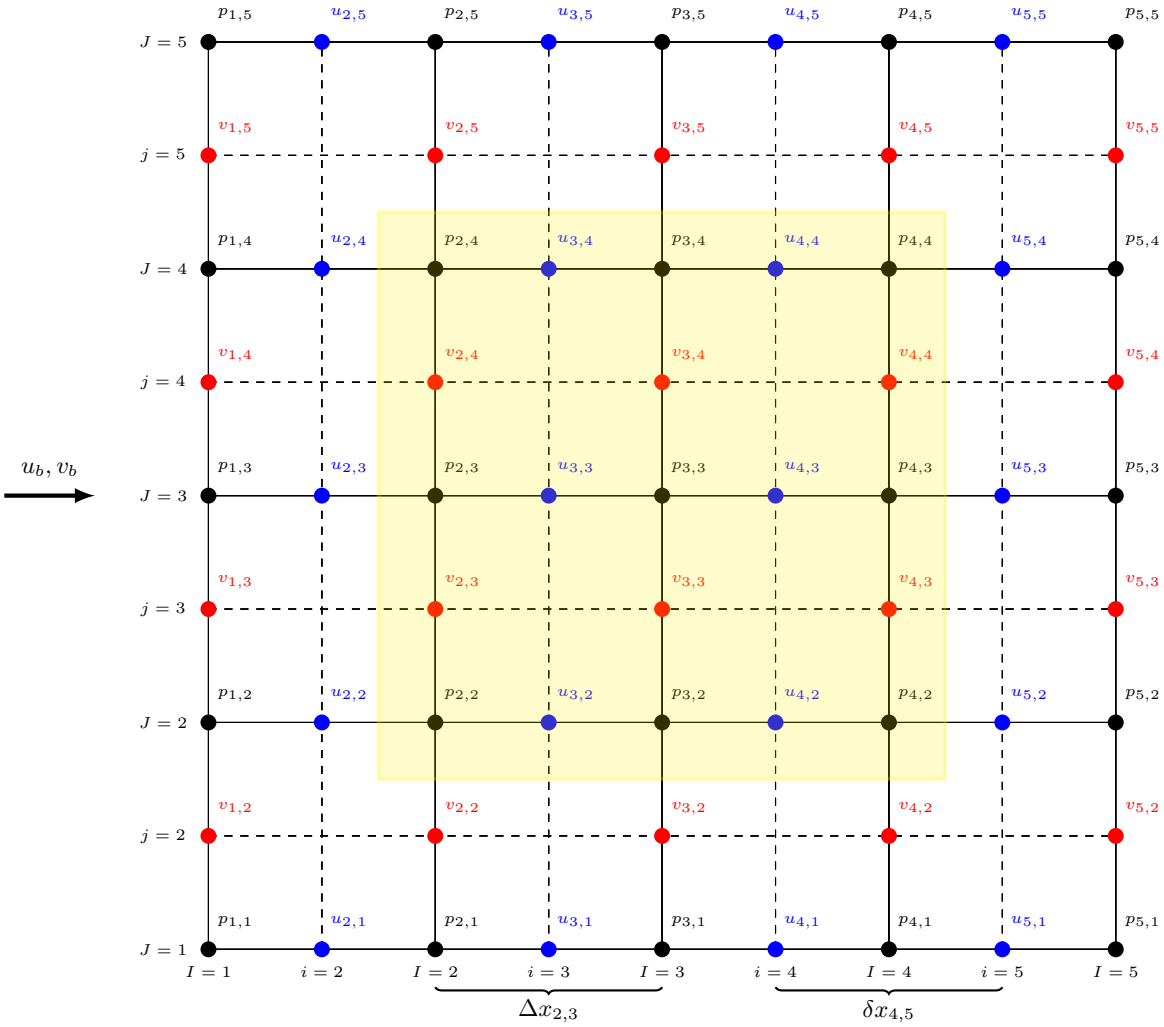


Figure 1: Example of a staggered grid with $N_x = N_y = 5$.

Note that in the program we have one extra column on the west side for the u-velocity components and one extra row on the south side for the v-velocity component. In this way, i and j both begin at one in the program. The boundary conditions are such that $u_{2,J} = u_b$ and $v_{2,j} = v_b$ while on the other three boundaries zero gradient boundary conditions are imposed meaning

$$u_{i,1} = u_{i,2}, \qquad u_{N_x,J} = u_{N_x-1,J}, \qquad u_{i,N_y} = u_{i,N_y-1}$$
$$v_{I,1} = v_{I,2}, \qquad v_{N_x,j} = v_{N_x-1,j}, \qquad v_{I,N_y} = v_{I,N_y-1}$$

Using this example we can formulate the matrix equations (after neglecting the actuator and dynamical terms for simplicity) given in Eq. 2 as follows:

$$
\left[\begin{array}{ccc|ccc|cc|cc|cc|ccc|ccc|ccc}
a^{px}_{3,2} & -a^{nx}_{3,2} & 0 & -a^{ex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
-a^{sx}_{3,3} & a^{px}_{3,3} & -a^{nx}_{3,3} & 0 & -a^{ex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & -a^{sx}_{3,4} & a^{px}_{3,4} & 0 & 0 & -a^{ex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 \\
\hline
-a^{wx}_{4,2} & 0 & 0 & a^{px}_{4,2} & -a^{nx}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 \\
0 & -a^{wx}_{4,3} & 0 & -a^{sx}_{4,3} & a^{px}_{4,3} & -a^{nx}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 \\
0 & 0 & -a^{wx}_{4,4} & 0 & -a^{sx}_{4,4} & a^{px}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & a^{py}_{2,3} & -a^{ny}_{2,3} & -a^{ey}_{2,3} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -a^{sy}_{2,4} & a^{py}_{2,4} & 0 & -a^{ey}_{2,4} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{3,3} & 0 & a^{py}_{3,3} & -a^{ny}_{3,3} & -a^{ey}_{3,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{3,4} & -a^{sy}_{3,4} & a^{py}_{3,4} & 0 & -a^{ey}_{3,4} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{4,3} & 0 & a^{py}_{4,3} & -a^{ny}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{4,4} & -a^{sy}_{4,4} & a^{py}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} \\
\hline
-\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta y_{3,4} & 0 & 0 & 0 & 0 & \delta x_{2,3} & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & 0 & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta y_{2,3} & 0 & 0 & -\delta y_{2,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta y_{3,4} & 0 & 0 & -\delta y_{3,4} & 0 & 0 & 0 & \delta x_{3,4} & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta y_{4,5} & 0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & \delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]
\left(\begin{array}{c}
u_{3,2} \\ u_{3,3} \\ u_{3,4} \\ \hline u_{4,2} \\ u_{4,3} \\ u_{4,4} \\ \hline v_{2,3} \\ v_{2,4} \\ \hline v_{3,3} \\ v_{3,4} \\ \hline v_{4,3} \\ v_{4,4} \\ \hline p_{2,2} \\ p_{2,3} \\ p_{2,4} \\ \hline p_{3,2} \\ p_{3,3} \\ p_{3,4} \\ \hline p_{4,2} \\ p_{4,3} \\ p_{4,4}
\end{array}\right)
=
\left(\begin{array}{c}
a^{sx}_{3,2}u_{3,1} + a^{wx}_{3,2}u_{2,2} \\ a^{wx}_{3,3}u_{2,3} \\ \hline a^{wx}_{3,4}u_{2,4} + a^{nx}_{3,4}u_{3,5} \\ \hline a^{sx}_{4,2}u_{4,1} + a^{ex}_{4,2}u_{5,2} \\ a^{ex}_{4,3}u_{5,3} \\ \hline a^{ex}_{4,4}u_{5,4} + a^{nx}_{4,4}u_{4,5} \\ \hline a^{sy}_{2,3}v_{2,2} + a^{wx}_{2,3}v_{1,3} \\ a^{wy}_{2,4}v_{1,4} + a^{ny}_{2,4}v_{2,5} \\ \hline a^{sy}_{3,3}v_{3,2} \\ a^{ny}_{3,4}v_{3,5} \\ \hline a^{ey}_{4,3}v_{5,3} + a^{sy}_{4,3}v_{4,2} \\ a^{ey}_{4,4}v_{5,4} + a^{ny}_{4,4}v_{4,5} \\ \hline -\delta y_{2,3}u_{2,2} - \delta x_{2,3}v_{2,2} \\ -\delta y_{3,4}u_{2,3} \\ \hline -\delta y_{4,5}u_{2,4} + \delta x_{2,3}v_{2,5} \\ \hline -\delta x_{3,4}v_{3,2} \\ 0 \\ \hline \delta x_{3,4}v_{3,5} \\ \hline \delta y_{2,3}u_{5,2} - \delta x_{4,5}v_{4,2} \\ \delta y_{3,4}u_{5,3} \\ \delta y_{4,5}u_{5,4} + \delta x_{4,5}v_{4,5}
\end{array}\right)
$$

After applying the boundary conditions we get:

$$
\left[\begin{array}{ccc|ccc|cc|cc|cc|ccc|ccc|ccc}
a^{px}_{3,2}-a^{sx}_{3,2} & -a^{nx}_{3,2} & 0 & -a^{ex}_{3,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
-a^{sx}_{3,3} & a^{px}_{3,3} & -a^{nx}_{3,3} & 0 & -a^{ex}_{3,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & -a^{sx}_{3,4} & a^{px}_{3,4}-a^{nx}_{3,4} & 0 & 0 & -a^{ex}_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 \\
\hline
-a^{wx}_{4,2} & 0 & 0 & a^{px}_{4,2}- & -a^{nx}_{4,2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 \\
 & & & a^{sx}_{4,2}-a^{ex}_{4,2} & & & & & & & & & & & & & & & & & \\
0 & -a^{wx}_{4,3} & 0 & -a^{sx}_{4,3} & a^{px}_{4,3}-a^{ex}_{4,3} & -a^{nx}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 \\
0 & 0 & -a^{wx}_{4,4} & 0 & -a^{sx}_{4,4} & a^{px}_{4,4}- & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} \\
 & & & & & a^{ex}_{4,4}-a^{nx}_{4,4} & & & & & & & & & & & & & & & \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & a^{py}_{2,3}-a^{sy}_{2,3} & -a^{ny}_{2,3} & -a^{ey}_{2,3} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -a^{sy}_{2,4} & a^{py}_{2,4}- & 0 & -a^{ey}_{2,4} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 \\
 & & & & & & & a^{ny}_{2,4} & & & & & & & & & & & & & \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{3,3} & 0 & a^{py}_{3,3}- & -a^{ny}_{3,3} & -a^{ey}_{3,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 & 0 \\
 & & & & & & & & a^{sy}_{3,3} & & & & & & & & & & & & \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{3,4} & -a^{sy}_{3,4} & a^{py}_{3,4}- & 0 & -a^{ey}_{3,4} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 \\
 & & & & & & & & & a^{ny}_{3,4} & & & & & & & & & & & \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{4,3} & 0 & a^{py}_{4,3}- & -a^{ny}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} & 0 \\
 & & & & & & & & & & a^{ey}_{4,3}- & & & & & & & & & & \\
 & & & & & & & & & & a^{sy}_{4,3} & & & & & & & & & & \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -a^{wy}_{4,4} & -a^{sy}_{4,4} & a^{py}_{4,4}- & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} \\
 & & & & & & & & & & & a^{ey}_{4,4}- & & & & & & & & & \\
 & & & & & & & & & & & a^{ny}_{4,4} & & & & & & & & & \\
\hline
-\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta y_{3,4} & 0 & 0 & 0 & 0 & \delta x_{2,3} & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
\delta y_{2,3} & 0 & 0 & -\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta y_{3,4} & 0 & 0 & -\delta y_{3,4} & 0 & 0 & 0 & \delta x_{3,4} & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta y_{4,5} & 0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}\right]
\left(\begin{array}{c}
u_{3,2} \\ u_{3,3} \\ u_{3,4} \\ \hline u_{4,2} \\ u_{4,3} \\ u_{4,4} \\ \hline v_{2,3} \\ v_{2,4} \\ \hline v_{3,3} \\ v_{3,4} \\ \hline v_{4,3} \\ v_{4,4} \\ \hline p_{2,2} \\ p_{2,3} \\ p_{2,4} \\ \hline p_{3,2} \\ p_{3,3} \\ p_{3,4} \\ \hline p_{4,2} \\ p_{4,3} \\ p_{4,4}
\end{array}\right)
=
\left(\begin{array}{c}
a^{wx}_{3,2}u_b \\ a^{wx}_{3,3}u_b \\ a^{wx}_{3,4}u_b \\ \hline 0 \\ 0 \\ 0 \\ \hline a^{wx}_{2,3}v_b \\ a^{wy}_{2,4}v_b \\ \hline 0 \\ 0 \\ \hline 0 \\ 0 \\ \hline -\delta y_{2,3}u_b \\ -\delta y_{3,4}u_b \\ -\delta y_{4,5}u_b \\ \hline 0 \\ 0 \\ 0 \\ \hline 0 \\ 0 \\ 0
\end{array}\right)
$$

1. Linearisation

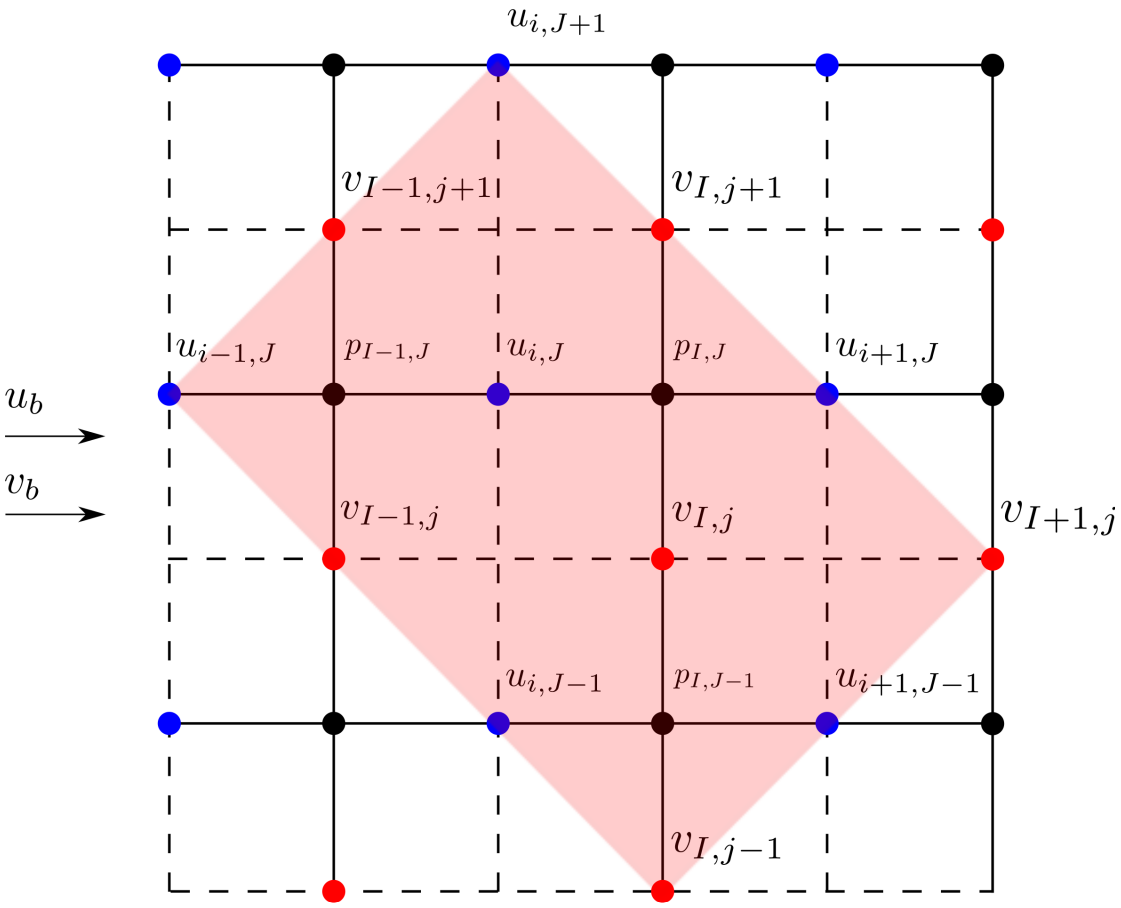


Figure 2: One cell in the grid for which we define one x-, y-momentum and one continuity equation (used for derivatives).

The discretization method used to discretize the 2D NS equations using the Hybrid differencing scheme. After discretizing, we obtain the equations as given in Table 1. We have the following dependence:

$a^{wx}(u_{i-1,J}, u_{i,J})$, $a^{ex}(u_{i,J}, u_{i+1,J})$, $a^{nx}(v_{I-1,j+1}, v_{I,j+1}, u_{i,J}, u_{i,J+1})$, $a^{sx}(v_{I-1,j}, v_{I,j}, u_{i,J-1}, u_{i,J})$,
$a^{px}(u_{i-1,J}, u_{i,J-1}, u_{i,J}, u_{i,J+1}, u_{i+1,J}, v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1})$,
$a^{wy}(u_{i,J-1}, u_{i,J})$, $a^{ey}(u_{i+1,J-1}, u_{i+1,J})$, $a^{ny}(v_{I,j+1}, v_{I,j})$, $a^{sy}(v_{I,j}, v_{I,j-1})$,
$a^{py}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1})$

According to Fig. 2 we can write down the following equations:

$$\frac{\mathrm{d}}{\mathrm{d}t}u_{i,J} = a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J}) + S^x_{i,J}, \quad \forall (I,i) \in \{3, N_x - 1\}, (J,j) \in \{2, N_y - 1\}, \quad \text{x-momentum}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}v_{I,j} = a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J}) + S^y_{I,j}, \quad \forall (I,i) \in \{2, N_x - 1\}, (J,j) \in \{3, N_y - 1\}, \quad \text{y-momentum}$$
$$0 = c_1 (u_{i+1,J} - u_{i,J}) + c_2 (v_{I,j+1} - v_{I,j}), \quad \forall (I,i) \in \{2, N_x - 1\}, (J,j) \in \{2, N_y - 1\}, \text{continuity}$$

with

$$a_x(x) = -(a^{px} + T^{nx} + T^{sx})u_{i,J} + a^{wx}u_{i-1,J} + a^{ex}u_{i+1,J} + (a^{nx} + T^{nx})u_{i,J+1} + (a^{sx} + T^{sx})u_{i,J-1}$$
$$a_y(x) = -a^{py}v_{I,j} + a^{wy}v_{I-1,j} + a^{ey}v_{I+1,j} + a^{ny}v_{I,j+1} + a^{sy}v_{I,j-1}$$

and a_{xc}, a_{yc}, c_1, c_2 constant coefficients depending on the grid (see Table 1). Furthermore we have that all a coefficients depend on certain velocity components. We have

$$a^{wx}(u_{i-1,J}, u_{i,J}),\ a^{ex}(u_{i,J}, u_{i+1,J}),\ a^{nx}(v_{I-1,j+1}, v_{I,j+1}, u_{i,J}, u_{i,J+1}),\ a^{sx}(v_{I-1,j}, v_{I,j}, u_{i,J-1}, u_{i,J}),$$
$$a^{px}(u_{i-1,J}, u_{i,J-1}, u_{i,J}, u_{i,J+1}, u_{i+1,J}, v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1}),$$
$$a^{wy}(u_{i,J-1}, u_{i,J}),\ a^{ey}(u_{i+1,J-1}, u_{i+1,J}),\ a^{ny}(v_{I,j+1}, v_{I,j}),\ a^{sy}(v_{I,j}, v_{I,j-1}),$$
$$a^{py}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1})$$

The turbulence terms are defined as:

$$T^{nx} = c_n |u_{i,J+1} - u_{i,J}|, \qquad T^{sx} = c_s |u_{i,J} - u_{i,J-1}| \tag{3}$$

Accordingly we can define a reduced state x_r, a state x and input w as

$$x_r = \begin{pmatrix} u_{i,J} & v_{I,j} \end{pmatrix}^T$$
$$x = \left(u_{i-1,J}\ u_{i,J-1}\ u_{i,J}\ u_{i,J+1}\ u_{i+1,J-1}\ u_{i+1,J} \middle| v_{I-1,j}\ v_{I-1,j+1}\ v_{I,j-1}\ v_{I,j}\ v_{I,j+1}\ v_{I+1,j+1} \middle| p_{I-1,J}\ p_{I,J-1}\ p_{I,J} \right)^T$$
$$w = \begin{pmatrix} S^x_{i,J} & S^y_{I,j} \end{pmatrix}^T$$

The general formulation for one cell is

$$\begin{aligned} \dot{x}_r &= f(x) + g(w) \\ 0 &= h(x) \end{aligned} \tag{4}$$

with

$$f(x) = \begin{pmatrix} a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J}) \\ a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J}) \end{pmatrix}, \quad g(w) = \begin{pmatrix} S^x_{i,J} \\ S^y_{I,j} \end{pmatrix}, \quad h(x) = c_1 (u_{i+1,J} - u_{i,J}) + c_2 (v_{I,j+1} - v_{I,j})$$
$$= \begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix}$$

If we linearise we get the following linearised model for one cell:

$$\dot{x}_r = \underbrace{\left.\frac{\partial f(x)}{\partial x}\right|_{x_0}}_{A} \delta x + \underbrace{\left.\frac{\partial g(w)}{\partial w}\right|_{w_0}}_{B} \delta w, \qquad 0 = h(\delta x) \tag{5}$$

with $\delta x, \delta w$ the deviation variables and x_0 and w_0 the points of linearisation. We have as deviation state variable:

$$\delta x = \Big(\delta u_{i-1,J}\ \delta u_{i,J-1}\ \delta u_{i,J}\ \delta u_{i,J+1}\ \delta u_{i+1,J-1}\ \delta u_{i+1,J} \Big| \delta v_{I-1,j}\ \delta v_{I-1,j+1}\ \delta v_{I,j-1}\ \delta v_{I,j}\ \delta v_{I,j+1}\ \delta v_{I+1,j} \Big| \cdots$$
$$\cdots\ \delta p_{I-1,J}\ \delta p_{I,J-1}\ \delta p_{I,J} \Big)$$

Furthermore we have

$$\frac{\partial f(x)}{\partial x} =$$
$$\begin{pmatrix} f_1(x) \\ f_2(x) \end{pmatrix} \Big(\frac{\partial}{\partial u_{i-1,J}}\ \frac{\partial}{\partial u_{i,J-1}}\ \frac{\partial}{\partial u_{i,J}}\ \frac{\partial}{\partial u_{i,J+1}}\ \frac{\partial}{\partial u_{i+1,J-1}}\ \frac{\partial}{\partial u_{i+1,J}} \Big| \frac{\partial}{\partial v_{I-1,j}}\ \frac{\partial}{\partial v_{I-1,j+1}}\ \frac{\partial}{\partial v_{I,j-1}}\ \frac{\partial}{\partial v_{I,j}}\ \frac{\partial}{\partial v_{I,j+1}}\ \frac{\partial}{\partial v_{I+1,j+1}} \Big| \cdots$$
$$\cdots\ \frac{\partial}{\partial p_{I-1,J}}\quad \frac{\partial}{\partial p_{I,J-1}}\quad \frac{\partial}{\partial p_{I,J}} \Big)$$
$$\frac{\partial g(w)}{\partial w} = \begin{pmatrix} g_1(w) \\ g_2(w) \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial w_1} & \frac{\partial}{\partial w_2} \end{pmatrix}$$

In Eq. 5 we have a linearised model for one cell. If we lift the state variable to a deviation state variable $\delta\tilde{x}$ we can obtain a set of equations containing the linearised models of all the cells in the field.

In the following subsections we will compute the partial derivatives of first the x-momentum and then the y-momentum equation.

1.1. x-momentum

We have the spatially discretised x-momentum equation

$$\frac{\mathrm{d}}{\mathrm{d}t}u_{i,J} = \underbrace{a_x(x) + a_{xc}(p_{I-1,J} - p_{I,J})}_{f_1(x)} + S^x_{i,J} \tag{6}$$

with

$$a_x(x) = -(a^{px} + T^{nx} + T^{sx})u_{i,J} + a^{wx}u_{i-1,J} + a^{ex}u_{i+1,J} + (a^{nx} + T^{nx})u_{i,J+1} + (a^{sx} + T^{sx})u_{i,J-1}$$

We have the following dependency of the coefficients:

$a^{wx}(u_{i-1,J}, u_{i,J})$, $a^{ex}(u_{i,J}, u_{i+1,J})$, $a^{nx}(v_{I-1,j+1}, v_{I,j+1}, u_{i,J}, u_{i,J+1})$, $a^{sx}(v_{I-1,j}, v_{I,j}, u_{i,J-1}, u_{i,J})$, $a^{px}(u_{i-1,J}, u_{i,J-1}, u_{i,J}, u_{i,J+1}, u_{i+1,J}, v_{I-1,j}, v_{I-1,j+1}, v_{I,j}, v_{I,j+1})$,

When using Eq. 5 we get:

$$\frac{\partial f_1(x)}{\partial x} = \Bigg(\frac{\partial f_1(x)}{\partial u_{i-1,J}} \; \frac{\partial f_1(x)}{\partial u_{i,J-1}} \; \frac{\partial f_1(x)}{\partial u_{i,J}} \; \frac{\partial f_1(x)}{\partial u_{i,J+1}} \; \frac{\partial f_1(x)}{\partial u_{i+1,J-1}} \; \frac{\partial f_1(x)}{\partial u_{i+1,J}} \Big| \frac{\partial f_1(x)}{\partial v_{I-1,j}} \; \frac{\partial f_1(x)}{\partial v_{I-1,j+1}} \; \frac{\partial f_1(x)}{\partial v_{I,j-1}} \; \frac{\partial f_1(x)}{\partial v_{I,j}} \; \frac{\partial f_1(x)}{\partial v_{I,j+1}} \; \frac{\partial f_1(x)}{\partial v_{I+1,j}} \Big| \cdots$$
$$\cdots \; \frac{\partial f_1(x)}{\partial p_{I-1,J}} \; \frac{\partial f_1(x)}{\partial p_{I,J-1}} \; \frac{\partial f_1(x)}{\partial p_{I,J}} \Bigg)$$

with

$$
\begin{aligned}
\frac{\partial f_1(x)}{\partial u_{i-1,J}} &= -\frac{\partial a^{px}}{\partial u_{i-1,J}} u_{i,J} + a^{wx} + \frac{\partial a^{wx}}{\partial u_{i-1,J}} u_{i-1,J}, \ (= \delta a^{wx}) \\
\frac{\partial f_1(x)}{\partial u_{i,J-1}} &= -\frac{\partial T^{sx}}{\partial u_{i,J-1}} u_{i,J} + a^{sx} + \frac{\partial T^{sx}}{\partial u_{i,J-1}} u_{i,J-1} + T^{sx}, \ (= \delta a^{sx}) \\
\frac{\partial f_1(x)}{\partial u_{i,J}} &= -\frac{\partial a^{px}}{\partial u_{i,J}} u_{i,J} - a^{px} + \frac{\partial a^{wx}}{\partial u_{i,J}} u_{i-1,J} + \frac{\partial a^{ex}}{\partial u_{i,J}} u_{i+1,J} + \\
&\quad -\frac{\partial T^{nx}}{\partial u_{i,J}} u_{i,J} - T^{nx} - \frac{\partial T^{sx}}{\partial u_{i,J}} u_{i,J} - T^{sx} + \frac{\partial T^{nx}}{\partial u_{i,J}} u_{i,J+1} + \frac{\partial T^{sx}}{\partial u_{i,J}} u_{i,J-1} \ (= \delta a^{px}) \\
\frac{\partial f_1(x)}{\partial u_{i,J+1}} &= -\frac{\partial T^{nx}}{\partial u_{i,J+1}} u_{i,J} + a^{nx} + \frac{\partial T^{nx}}{\partial u_{i,J+1}} u_{i,J+1} + T^{nx}, \ (= \delta a^{nx}) \\
\frac{\partial f_1(x)}{\partial u_{i+1,J-1}} &= 0, \\
\frac{\partial f_1(x)}{\partial u_{i+1,J}} &= -\frac{\partial a^{px}}{\partial u_{i+1,J}} u_{i,J} + \frac{\partial a^{ex}}{\partial u_{i+1,J}} u_{i+1,J} + a^{ex}, \ (= \delta a^{ex}) \\
\frac{\partial f_1(x)}{\partial v_{I-1,j}} &= -\frac{\partial a^{P_x}}{\partial v_{I-1,j}} u_{i,J} + \frac{\partial a^{S_x}}{\partial v_{I-1,j}} u_{i,J-1}, \ (= \delta a^{swx}) \\
\frac{\partial f_1(x)}{\partial v_{I-1,j+1}} &= -\frac{\partial a^{px}}{\partial v_{I-1,j+1}} u_{i,J} + \frac{\partial a^{nx}}{\partial v_{I-1,j+1}} u_{i,J+1}, \ (= \delta a^{nwx}) \\
\frac{\partial f_1(x)}{\partial v_{I,j-1}} &= 0, \\
\frac{\partial f_1(x)}{\partial v_{I,j}} &= -\frac{\partial a^{px}}{\partial v_{I,j}} u_{i,J} + \frac{\partial a^{sx}}{\partial v_{I,j}} u_{i,J-1}, \ (= \delta a^{sex}) \\
\frac{\partial f_1(x)}{\partial v_{I,j+1}} &= -\frac{\partial a^{px}}{\partial v_{I,j+1}} u_{i,J} + \frac{\partial a^{nx}}{\partial v_{I,j+1}} u_{i,J+1}, \ (= \delta a^{nex}) \\
\frac{\partial f_1(x)}{\partial v_{I+1,j}} &= 0, \\
\frac{\partial f_1(x)}{\partial p_{I-1,J}} &= a_{xc}, \\
\frac{\partial f_1(x)}{\partial p_{I,J-1}} &= 0, \\
\frac{\partial f_1(x)}{\partial p_{I,J}} &= -a_{xc},
\end{aligned}
$$

Note that all the partials are evaluated at an equilibrium.

1.2. y-momentum

We have the spatially discretised y-momentum equation

$$\frac{\mathrm{d}}{\mathrm{d}t}v_{I,j} = \underbrace{a_y(x) + a_{yc}(p_{I,J-1} - p_{I,J})}_{f_2(x)} + S^y_{I,j} \tag{7}$$

with

$$a_y(x) = -a^{py}v_{I,j} + a^{wy}v_{I-1,j} + a^{ey}v_{I+1,j} + a^{ny}v_{I,j+1} + a^{sy}v_{I,j-1} \tag{8}$$

We have the following dependency of the coefficients:

$$a^{wy}(u_{i,J-1}, u_{i,J}),\ a^{ey}(u_{i+1,J-1}, u_{i+1,J}),\ a^{ny}(v_{I,j+1}, v_{I,j}),\ a^{sy}(v_{I,j}, v_{I,j-1}),$$
$$a^{py}(u_{i,J-1}, u_{i,J}, u_{i+1,J-1}, u_{i+1,J}, v_{I,j-1}, v_{I,j}, v_{I,j+1})$$

When using Eq. 5 we get:

$$\frac{\partial f_2(x)}{\partial x} = \Bigg(\frac{\partial f_2(x)}{\partial u_{i-1,J}} \ \frac{\partial f_2(x)}{\partial u_{i,J-1}} \ \frac{\partial f_2(x)}{\partial u_{i,J}} \ \frac{\partial f_2(x)}{\partial u_{i,J+1}} \ \frac{\partial f_2(x)}{\partial u_{i+1,J-1}} \ \frac{\partial f_2(x)}{\partial u_{i+1,J}} \Big| \frac{\partial f_2(x)}{\partial v_{I-1,j}} \ \frac{\partial f_2(x)}{\partial v_{I-1,j+1}} \ \frac{\partial f_2(x)}{\partial v_{I,j-1}} \ \frac{\partial f_2(x)}{\partial v_{I,j}} \ \frac{\partial f_2(x)}{\partial v_{I,j+1}} \ \frac{\partial f_2(x)}{\partial v_{I+1,j}} \Big| \cdots$$
$$\cdots \ \frac{\partial f_2(x)}{\partial p_{I-1,J}} \ \frac{\partial f_2(x)}{\partial p_{I,J-1}} \ \frac{\partial f_2(x)}{\partial p_{I,J}} \Bigg)$$

with

$$
\begin{aligned}
\frac{\partial f_2(x)}{\partial u_{i-1,J}} &= 0, \\
\frac{\partial f_2(x)}{\partial u_{i,J-1}} &= -\frac{\partial a^{py}}{\partial u_{i,J-1}} v_{I,j} + \frac{\partial a^{wy}}{\partial u_{i,J-1}} v_{I-1,j}, \ (= \delta a^{swy}) \\
\frac{\partial f_2(x)}{\partial u_{i,J}} &= -\frac{\partial a^{py}}{\partial u_{i,J}} v_{I,j} + \frac{\partial a^{wy}}{\partial u_{i,J}} v_{I-1,j}, \ (= \delta a^{nwy}) \\
\frac{\partial f_2(x)}{\partial u_{i,J+1}} &= 0, \\
\frac{\partial f_2(x)}{\partial u_{i+1,J-1}} &= -\frac{\partial a^{py}}{\partial u_{i+1,J-1}} v_{I,j} + \frac{\partial a^{ey}}{\partial u_{i+1,J-1}} v_{I+1,j}, \ (= \delta a^{sey}) \\
\frac{\partial f_2(x)}{\partial u_{i+1,J}} &= -\frac{\partial a^{py}}{\partial u_{i+1,J}} v_{I,j} + \frac{\partial a^{ey}}{\partial u_{i+1,J}} v_{I+1,j}, \ (= \delta a^{ney}) \\
\frac{\partial f_2(x)}{\partial v_{I-1,j}} &= a^{wy}, \ (= \delta a^{wy}) \\
\frac{\partial f_2(x)}{\partial v_{I-1,j+1}} &= 0, \\
\frac{\partial f_2(x)}{\partial v_{I,j-1}} &= -\frac{\partial a^{py}}{\partial v_{I,j-1}} v_{I,j} + \frac{\partial a^{sy}}{\partial v_{I,j-1}} v_{I,j-1} + a^{sy}, \ (= \delta a^{sy}) \\
\frac{\partial f_2(x)}{\partial v_{I,j}} &= -\frac{\partial a^{py}}{\partial v_{I,j}} v_{I,j} - a^{py} + \frac{\partial a^{ny}}{\partial v_{I,j}} v_{I,j+1} + \frac{\partial a^{sy}}{\partial v_{I,j}} v_{I,j-1}, \ (= \delta a^{py}) \\
\frac{\partial f_2(x)}{\partial v_{I,j+1}} &= -\frac{\partial a^{py}}{\partial v_{I,j+1}} v_{I,j} + \frac{\partial a^{ny}}{\partial v_{I,j+1}} v_{I,j+1} + a^{ny}, \ (= \delta a^{ny}) \\
\frac{\partial f_2(x)}{\partial v_{I+1,j}} &= a^{ey}, \ (= \delta a^{ey}), \\
\frac{\partial f_2(x)}{\partial p_{I-1,J}} &= 0, \\
\frac{\partial f_2(x)}{\partial p_{I,J-1}} &= a_{yc}, \\
\frac{\partial f_2(x)}{\partial p_{I,J}} &= -a_{yc},
\end{aligned}
$$

Note that all the partials are evaluated at an equilibrium.

1.3. Continuity equation
The linearised continuity equation is defined as:

$$\begin{aligned} 0 &= c_1 \left(\delta u_{i+1,J} - \delta u_{i,J}\right) + c_2 \left(\delta v_{I,j+1} - \delta v_{I,j}\right), \\ &= h(\delta x) \end{aligned}$$

2. Temporal discretization and lifting the state vector
Finally we can obtain, after temporal discretisation and lifting the state vector to $\delta\tilde{x}$, the following linearised equation for the complete field:

$$\begin{pmatrix} A_1 & A_2 & B_1 \\ A_3 & A_4 & B_2 \\ B_1^T & B_2^T & 0 \end{pmatrix} \delta\tilde{x}_{k+1} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \tag{9}$$

The used temporal differentiation method is the implicit one and we define:

$$\begin{aligned} \rho\frac{\partial u_{i,J}}{\partial t} &= \frac{\rho\Delta V}{t}(u_{i,J} - u_{i,J}^0) \\ \rho\frac{\partial v_{I,j}}{\partial t} &= \frac{\rho\Delta V}{t}(v_{I,j} - v_{I,j}^0) \end{aligned}$$

Example continued (without turbulence)

$$
\begin{bmatrix}
\delta a^{px}_{3,2} & -\delta a^{nx}_{3,2} & 0 & -\delta a^{ex}_{3,2} & 0 & 0 & -\delta a^{nwx}_{3,2} & 0 & -\delta a^{nex}_{3,2} & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
-\delta a^{sx}_{3,3} & \delta a^{px}_{3,3} & -\delta a^{nx}_{3,3} & 0 & -\delta a^{ex}_{3,3} & 0 & -\delta a^{swx}_{3,3} & -\delta a^{nwx}_{3,3} & -\delta a^{sex}_{3,3} & -\delta a^{nex}_{3,2} & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & -\delta a^{sx}_{3,4} & \delta a^{px}_{3,4} & 0 & 0 & -\delta a^{ex}_{3,4} & 0 & -\delta a^{swx}_{3,4} & 0 & -\delta a^{sex}_{3,4} & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 \\
-\delta a^{wx}_{4,2} & 0 & 0 & \delta a^{px}_{4,2} & -\delta a^{nx}_{4,2} & 0 & 0 & 0 & -\delta a^{nwx}_{4,2} & 0 & -\delta a^{nex}_{4,2} & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 \\
0 & -\delta a^{wx}_{4,3} & 0 & -\delta a^{sx}_{4,3} & \delta a^{px}_{4,3} & -\delta a^{nx}_{4,3} & 0 & 0 & -\delta a^{swx}_{4,3} & -\delta a^{nwx}_{4,3} & -\delta a^{nex}_{4,3} & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 \\
0 & 0 & -\delta a^{wx}_{4,4} & 0 & -\delta a^{sx}_{4,4} & \delta a^{px}_{4,4} & 0 & 0 & 0 & -\delta a^{swx}_{4,4} & 0 & -\delta a^{sex}_{4,4} & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} \\
-\delta a^{sey}_{2,3} & -\delta a^{ney}_{2,3} & 0 & 0 & 0 & 0 & \delta a^{py}_{2,3} & -\delta a^{ny}_{2,3} & -\delta a^{ey}_{2,3} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta a^{sey}_{2,4} & -\delta a^{ney}_{2,4} & 0 & 0 & 0 & -\delta a^{sy}_{2,4} & \delta a^{py}_{2,4} & 0 & -\delta a^{ey}_{2,4} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\delta a^{swy}_{3,3} & -\delta a^{nwy}_{3,3} & 0 & -\delta a^{sey}_{3,3} & -\delta a^{ney}_{3,3} & 0 & -\delta a^{wy}_{3,3} & 0 & \delta a^{py}_{3,3} & -\delta a^{ny}_{3,3} & -\delta a^{ey}_{3,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 & 0 \\
0 & -\delta a^{swy}_{3,4} & -\delta a^{nwy}_{3,4} & 0 & -\delta a^{sey}_{3,4} & -\delta a^{ney}_{3,4} & 0 & -\delta a^{wy}_{3,4} & -\delta a^{sy}_{3,4} & \delta a^{py}_{3,4} & 0 & -\delta a^{ey}_{3,4} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 \\
0 & 0 & 0 & -\delta a^{swy}_{4,3} & -\delta a^{nwy}_{4,3} & 0 & 0 & 0 & -\delta a^{wy}_{4,3} & 0 & \delta a^{py}_{4,3} & -\delta a^{ny}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} & 0 \\
0 & 0 & 0 & 0 & -\delta a^{swy}_{4,4} & -\delta a^{nwy}_{4,4} & 0 & 0 & 0 & -\delta a^{wy}_{4,4} & -\delta a^{sy}_{4,4} & \delta a^{py}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} \\
-\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta y_{3,4} & 0 & 0 & 0 & 0 & \delta x_{2,3} & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & 0 & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\delta y_{2,3} & 0 & 0 & -\delta y_{2,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta y_{3,4} & 0 & 0 & -\delta y_{3,4} & 0 & 0 & 0 & \delta x_{3,4} & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta y_{4,5} & 0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & \delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\begin{pmatrix}
\delta u_{3,2} \\ \delta u_{3,3} \\ \delta u_{3,4} \\ \delta u_{4,2} \\ \delta u_{4,3} \\ \delta u_{4,4} \\ \delta v_{2,3} \\ \delta v_{2,4} \\ \delta v_{3,3} \\ \delta v_{3,4} \\ \delta v_{4,3} \\ \delta v_{4,4} \\ \delta p_{2,2} \\ \delta p_{2,3} \\ \delta p_{2,4} \\ \delta p_{3,2} \\ \delta p_{3,3} \\ \delta p_{3,4} \\ \delta p_{4,2} \\ \delta p_{4,3} \\ \delta p_{4,4}
\end{pmatrix}
=
\begin{pmatrix}
\delta a^{sx}_{3,2} u_{3,1} + \delta a^{wx}_{3,2} u_{2,2} + \delta a^{swx}_{3,2} v_{2,2} + \delta a^{sex}_{3,2} v_{3,2} \\
\delta a^{wx}_{3,3} u_{2,3} \\
\delta a^{wx}_{3,4} u_{2,4} + \delta a^{nx}_{3,4} u_{3,5} + \delta a^{nwx}_{3,4} v_{2,5} + \delta a^{nex}_{3,4} v_{3,5} \\
\delta a^{sx}_{4,2} u_{4,1} + \delta a^{ex}_{4,2} u_{5,2} + \delta a^{swx}_{4,2} v_{3,2} + \delta a^{sex}_{4,2} v_{4,2} \\
\delta a^{ex}_{4,3} u_{5,3} \\
\delta a^{ex}_{4,4} u_{5,4} + \delta a^{nx}_{4,4} u_{4,5} + \delta a^{nwx}_{4,4} v_{3,5} + \delta a^{nex}_{4,4} v_{4,5} \\
\delta a^{sy}_{2,3} v_{2,2} + \delta a^{wy}_{2,3} v_{1,3} + \delta a^{swy}_{2,3} u_{2,2} + \delta a^{nwy}_{2,3} u_{2,3} \\
\delta a^{wy}_{2,4} v_{1,4} + \delta a^{ny}_{2,4} v_{2,5} + \delta a^{swy}_{2,4} u_{2,3} + \delta a^{nwy}_{2,4} u_{2,4} \\
\delta a^{sy}_{3,3} v_{3,2} \\
\delta a^{ny}_{3,4} v_{3,5} \\
\delta a^{ey}_{4,3} v_{5,3} + \delta a^{sy}_{4,3} v_{4,2} + \delta a^{sey}_{4,3} u_{5,2} + \delta a^{ney}_{4,3} u_{5,3} \\
\delta a^{ey}_{4,4} v_{5,4} + \delta a^{ny}_{4,4} v_{4,5} + \delta a^{sey}_{4,4} u_{5,3} + \delta a^{ney}_{4,4} u_{5,4} \\
-\delta y_{2,3} u_{2,2} - \delta x_{2,3} v_{2,2} \\
-\delta y_{3,4} u_{2,3} \\
-\delta y_{4,5} u_{2,4} + \delta x_{2,3} v_{2,5} \\
-\delta x_{3,4} v_{3,2} \\
0 \\
\delta x_{3,4} v_{3,5} \\
\delta y_{2,3} u_{5,2} - \delta x_{4,5} v_{4,2} \\
\delta y_{3,4} u_{5,3} \\
\delta y_{4,5} u_{5,4} + \delta x_{4,5} v_{4,5}
\end{pmatrix}
$$

After applying boundary conditions:

$$
\begin{bmatrix}
\delta a^{px}_{3,2} - \delta a^{sx}_{3,2} & -\delta a^{nx}_{3,2} & 0 & -\delta a^{ex}_{3,2} & 0 & 0 & -\delta a^{nwx}_{3,2} - \delta a^{swx}_{3,2} & 0 & -\delta a^{nex}_{3,2} - \delta a^{sex}_{3,2} & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 \\
-\delta a^{sx}_{3,3} & \delta a^{px}_{3,3} & -\delta a^{nx}_{3,3} & 0 & -\delta a^{ex}_{3,3} & 0 & -\delta a^{swx}_{3,3} & -\delta a^{nwx}_{3,3} & -\delta a^{sex}_{3,3} & -\delta a^{nex}_{3,2} & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 \\
0 & -\delta a^{sx}_{3,4} & \delta a^{px}_{3,4} - \delta a^{nx}_{3,4} & 0 & 0 & -\delta a^{ex}_{3,4} & 0 & -\delta a^{swx}_{3,4} - \delta a^{nwx}_{3,4} & 0 & -\delta a^{sex}_{3,4} - \delta a^{nex}_{3,4} & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 \\
-\delta a^{wx}_{4,2} & 0 & 0 & \delta a^{px}_{4,2} - \delta a^{sx}_{4,2} - \delta a^{ex}_{4,2} & -\delta a^{nx}_{4,2} & 0 & 0 & 0 & -\delta a^{nwx}_{4,2} - \delta a^{swx}_{4,2} & 0 & -\delta a^{nex}_{4,2} - \delta a^{sex}_{4,2} & 0 & 0 & 0 & 0 & -\delta y_{2,3} & 0 & 0 & \delta y_{2,3} & 0 & 0 \\
0 & -\delta a^{wx}_{4,3} & 0 & -\delta a^{sx}_{4,3} & \delta a^{px}_{4,3} - \delta a^{ex}_{4,3} & -\delta a^{nx}_{4,3} & 0 & 0 & -\delta a^{nwx}_{4,3} & -\delta a^{nex}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta y_{3,4} & 0 & 0 & \delta y_{3,4} & 0 \\
0 & 0 & -\delta a^{wx}_{4,4} & 0 & -\delta a^{sx}_{4,4} & \delta a^{px}_{4,4} - \delta a^{ex}_{4,4} - \delta a^{nx}_{4,4} & 0 & 0 & 0 & -\delta a^{swx}_{4,4} - \delta a^{nwx}_{4,4} & 0 & -\delta a^{sex}_{4,4} - \delta a^{nex}_{4,4} & 0 & 0 & 0 & 0 & 0 & -\delta y_{4,5} & 0 & 0 & \delta y_{4,5} \\
-\delta a^{sey}_{2,3} & -\delta a^{ney}_{2,3} & 0 & 0 & 0 & 0 & \delta a^{py}_{2,3} - \delta a^{sy}_{2,3} & -\delta a^{ny}_{2,3} & -\delta a^{ey}_{2,3} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta a^{sey}_{2,4} & -\delta a^{ney}_{2,4} & 0 & 0 & 0 & -\delta a^{sy}_{2,4} & \delta a^{py}_{2,4} - \delta a^{ny}_{2,4} & 0 & -\delta a^{ey}_{2,4} & 0 & 0 & 0 & -\delta x_{2,3} & \delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 \\
-\delta a^{swy}_{3,3} & -\delta a^{nwy}_{3,3} & 0 & -\delta a^{sey}_{3,3} & -\delta a^{ney}_{3,3} & 0 & -\delta a^{wy}_{3,3} & 0 & \delta a^{py}_{3,3} - \delta a^{sy}_{3,3} & -\delta a^{ny}_{3,3} & -\delta a^{ey}_{3,3} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 & 0 \\
0 & -\delta a^{swy}_{3,4} & -\delta a^{nwy}_{3,4} & 0 & -\delta a^{sey}_{3,4} & -\delta a^{ney}_{3,4} & 0 & -\delta a^{wy}_{3,4} & -\delta a^{sy}_{3,4} & \delta a^{py}_{3,4} - \delta a^{ny}_{3,4} & 0 & -\delta a^{ey}_{3,4} & 0 & 0 & 0 & 0 & -\delta x_{3,4} & \delta x_{3,4} & 0 & 0 & 0 \\
0 & 0 & 0 & -\delta a^{swy}_{4,3} - \delta a^{sey}_{4,3} & -\delta a^{nwy}_{4,3} - \delta a^{ney}_{4,3} & 0 & 0 & 0 & -\delta a^{wy}_{4,3} & 0 & \delta a^{py}_{4,3} - \delta a^{ey}_{4,3} - \delta a^{sy}_{4,3} & -\delta a^{ny}_{4,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} & 0 \\
0 & 0 & 0 & 0 & -\delta a^{swy}_{4,4} - \delta a^{sey}_{4,4} & -\delta a^{nwy}_{4,4} - \delta a^{ney}_{4,4} & 0 & 0 & 0 & -\delta a^{wy}_{4,4} & -\delta a^{sy}_{4,4} & \delta a^{py}_{4,4} - \delta a^{ey}_{4,4} - \delta a^{ny}_{4,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & \delta x_{4,5} \\
-\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\delta y_{3,4} & 0 & 0 & 0 & 0 & \delta x_{2,3} & -\delta x_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\delta y_{2,3} & 0 & 0 & -\delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \delta y_{3,4} & 0 & 0 & -\delta y_{3,4} & 0 & 0 & 0 & \delta x_{3,4} & -\delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \delta y_{4,5} & 0 & 0 & -\delta y_{4,5} & 0 & 0 & 0 & \delta x_{3,4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \delta y_{2,3} & 0 & 0 & 0 & 0 & 0 & 0 & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \delta y_{3,4} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & -\delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \delta y_{4,5} & 0 & 0 & 0 & 0 & 0 & \delta x_{4,5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}
\delta u =
\begin{pmatrix}
\delta a^{wx}_{3,2} u_b \\ \delta a^{wx}_{3,3} u_b \\ \delta a^{wx}_{3,4} u_b \\ 0 \\ 0 \\ 0 \\ \delta a^{wy}_{2,3} v_b + (\delta a^{swy}_{2,3} + \delta a^{nwy}_{2,3}) u_b \\ \delta a^{wy}_{2,4} v_b + (\delta a^{swy}_{2,4} + \delta a^{nwy}_{2,4}) u_b \\ 0 \\ 0 \\ 0 \\ 0 \\ -\delta y_{2,3} u_b \\ -\delta y_{3,4} u_b \\ -\delta y_{4,5} u_b \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0
\end{pmatrix}
$$

2.1. Linearised model (other approach)
We have the following set of equations:

$$A(x_k)x_{k+1} = Qx_k + B_c(x_k) + S_m(x_k, \beta, \gamma) \tag{10}$$

with

$$x = \begin{pmatrix} u \\ v \\ p \end{pmatrix}, \quad u = \begin{pmatrix} u_{3,2} \\ u_{3,3} \\ \vdots \\ u_{3,N_y-1} \\ u_{4,2} \\ \vdots \\ u_{N_x-1,N_y-1} \end{pmatrix}, \quad v = \begin{pmatrix} v_{2,3} \\ v_{2,4} \\ \vdots \\ v_{2,N_y-1} \\ v_{3,3} \\ \vdots \\ v_{N_x-1,N_y-1} \end{pmatrix}, \quad p = \begin{pmatrix} p_{2,2} \\ p_{2,3} \\ \vdots \\ p_{2,N_y-1} \\ p_{3,2} \\ \vdots \\ p_{N_x-1,N_y-1} \end{pmatrix} \tag{11}$$

and

$$A(x_k) = \begin{bmatrix} A_x(x_k) & 0 & B_1 \\ 0 & A_y(x_k) & B_2 \\ B_1^T & B_2^T & 0 \end{bmatrix} \tag{12}$$

and

$$B_c(x_k) = \begin{pmatrix} a_{2,2}^{W_x}(u_b, u_{3,2})u_b \\ a_{2,3}^{W_x}(u_b, u_{3,3})u_b \\ \vdots \\ a_{2,N_y-1}^{W_x}(u_b, u_{3,N_y-1})u_b \\ 0 \\ \vdots \\ 0 \\ a_{1,3}^{W_y}(u_{3,1}, u_{3,2})v_b \\ a_{1,4}^{W_y}(u_{3,2}, u_{3,3})v_b \\ \vdots \\ a_{1,N_y-1}^{W_y}(u_{3,N_y-2}, u_{3,N_y-1})v_b \\ 0 \\ \vdots \\ 0 \\ c_1 u_b \\ c_2 u_b \\ \vdots \\ c_{N_y-1}u_b \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad S_m(x_k, \beta, \gamma) = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ -f(u_{r_1}, v_{r_1}, \beta_1)\cos(\gamma_1) \\ -f(u_{r_2}, v_{r_2}, \beta_1)\cos(\gamma_1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ -f(u_{r_3}, v_{r_3}, \beta_2)\cos(\gamma_2) \\ -f(u_{r_4}, v_{r_4}, \beta_2)\cos(\gamma_2) \\ 0 \\ \vdots \\ 0 \\ f(u_{r_1}, v_{r_1}, \beta_1)\sin(\gamma_1) \\ f(u_{r_2}, v_{r_2}, \beta_1)\sin(\gamma_1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ f(u_{r_3}, v_{r_3}, \beta_2)\sin(\gamma_2) \\ f(u_{r_4}, v_{r_4}, \beta_2)\sin(\gamma_2) \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Note that in this example we have two turbines and each turbine works on two cells. Linearising around a point $\bar{x}_k$ results in

$$A(\bar{x}_k)\delta x_{k+1} = \left(Q + \frac{\partial B_c(x_k)}{\partial x_k} - \frac{\partial A(x_k)}{\partial x_k}\bar{x}_{k+1} + \frac{\partial S_m}{\partial x_k}\right)\delta x_k + \frac{\partial S_m}{\partial \beta_k}\delta\beta_k + \frac{\partial S_m}{\partial \gamma_k}\delta\gamma_k$$

with deviation variables $\delta x_k, \delta\beta_k$ and $\delta\gamma_k$. Then we can define:

$$\frac{\partial B_c(x_k)}{\partial x_k} = B_c(x_k)\begin{pmatrix}\frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \cdots & \frac{\partial}{\partial x_N}\end{pmatrix} = \begin{pmatrix} \mathtt{diag}\left(\frac{\partial a^{W_x}_{2,2}}{\partial u_{3,2}}, \frac{\partial a^{W_x}_{2,3}}{\partial u_{3,3}}, \ldots, \frac{\partial a^{W_x}_{2,N_y-1}}{\partial u_{3,2}}\right) & 0 \\ 0 & 0 \\ \mathtt{diag}\left(\frac{\partial a^{W_y}_{1,3}}{\partial u_{3,2}}, \frac{\partial a^{W_y}_{1,4}}{\partial u_{3,3}}, \ldots, \frac{\partial a^{W_y}_{1,N_y-1}}{\partial u_{3,2}}\right) & 0 \\ 0 & 0 \end{pmatrix}$$

and

$$\frac{\partial S_m(x_k)}{\partial x_k} = S_m(x_k)\begin{pmatrix}\frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \cdots & \frac{\partial}{\partial x_N}\end{pmatrix} = \begin{pmatrix}
0 & \ldots & & & & & \ldots & & & & & \\
0 & \ldots & & & & 0 & \ldots & & & & & \\
0 & -\frac{\partial f\cos(\gamma_1)}{\partial u_{r_1}} & 0 & \ldots & & 0 & \ldots & 0 & -\frac{\partial f\cos(\gamma_1)}{\partial v_{r_1}} & \ldots & 0 & \\
0 & \ldots & -\frac{\partial f\cos(\gamma_1)}{\partial u_{r_2}} & 0 & \ldots & 0 & \ldots & & \ldots & -\frac{\partial f\cos(\gamma_1)}{\partial v_{r_2}} & 0 & \\
0 & \ldots & \ldots & & & 0 & \ldots & & \ldots & \ldots & 0 & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & & \\
0 & \ldots & 0 & 0 & \ldots & -\frac{\partial f\cos(\gamma_2)}{\partial u_{r_3}} & & & & & -\frac{\partial f\cos(\gamma_2)}{\partial v_{r_3}} & \\
0 & \ldots & & 0 & \ldots & & -\frac{\partial f\cos(\gamma_2)}{\partial u_{r_4}} & \ldots & 0 & & & -\frac{\partial f\cos(\gamma_2)}{\partial v_{r_4}} \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & & \\
0 & \ldots & & & & 0 & \ldots & & & & & \\
0 & \frac{\partial f\sin(\gamma_1)}{\partial u_{r_1}} & 0 & \ldots & & 0 & \ldots & 0 & \frac{\partial f\sin(\gamma_1)}{\partial v_{r_1}} & \ldots & 0 & \\
0 & \ldots & \frac{\partial f\sin(\gamma_1)}{\partial u_{r_2}} & 0 & \ldots & 0 & \ldots & & \ldots & \frac{\partial f\sin(\gamma_1)}{\partial v_{r_2}} & 0 & \\
0 & \ldots & \ldots & 0 & & 0 & \ldots & & \ldots & \ldots & 0 & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & & \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & & \\
0 & \ldots & 0 & 0 & \ldots & \frac{\partial f\sin(\gamma_2)}{\partial u_{r_3}} & & & & & \frac{\partial f\sin(\gamma_2)}{\partial v_{r_3}} & \\
0 & \ldots & & 0 & \ldots & & \frac{\partial f\sin(\gamma_2)}{\partial u_{r_4}} & \ldots & 0 & & & \frac{\partial f\sin(\gamma_2)}{\partial v_{r_4}} \\
0 & \ldots & \ldots & 0 & \ldots & & & \ldots & 0 & & &
\end{pmatrix}$$

with the derivatives as defined above. Furthermore we have

$$\frac{\partial S_m(\beta)}{\partial \beta} = S_m(\beta) \begin{pmatrix} \frac{\partial}{\partial \beta_1} & \frac{\partial}{\partial \beta_2} \end{pmatrix}$$

$$= \begin{pmatrix}
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
-\frac{\partial f(\beta_1)\cos(\gamma_1)}{\partial \beta_1} & 0 \\
-\frac{\partial f(\beta_1)\cos(\gamma_1)}{\partial \beta_1} & 0 \\
0 & 0 \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
0 & -\frac{\partial f(\beta_2)\cos(\gamma_2)}{\partial \beta_2} \\
0 & -\frac{\partial f(\beta_2)\cos(\gamma_2)}{\partial \beta_2} \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
\frac{\partial f(\beta_1)\sin(\gamma_1)}{\partial \beta_1} & 0 \\
\frac{\partial f(\beta_1)\sin(\gamma_1)}{\partial \beta_1} & 0 \\
0 & 0 \\
0 & 0 \\
\vdots & \vdots \\
0 & 0 \\
0 & \frac{\partial f(\beta_2)\sin(\gamma_2)}{\partial \beta_2} \\
0 & \frac{\partial f(\beta_2)\sin(\gamma_2)}{\partial \beta_2} \\
0 & 0 \\
\vdots & \vdots \\
0 & 0
\end{pmatrix}$$

and

$$\frac{\partial S_m(\gamma)}{\partial \gamma} = S_m(\gamma) \begin{pmatrix} \frac{\partial}{\partial \gamma_1} & \frac{\partial}{\partial \gamma_2} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ -\frac{\partial f(\gamma_1)\cos(\gamma_1)}{\partial \gamma_1} & 0 \\ -\frac{\partial f(\gamma_1)\cos(\gamma_1)}{\partial \gamma_1} & 0 \\ 0 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & -\frac{\partial f(\gamma_2)\cos(\gamma_2)}{\partial \gamma_2} \\ 0 & -\frac{\partial f(\gamma_2)\cos(\gamma_2)}{\partial \gamma_2} \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ \frac{\partial f(\gamma_1)\sin(\gamma_1)}{\partial \gamma_1} & 0 \\ \frac{\partial f(\gamma_1)\sin(\gamma_1)}{\partial \gamma_1} & 0 \\ 0 & 0 \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \\ 0 & \frac{\partial f(\gamma_2)\sin(\gamma_2)}{\partial \gamma_2} \\ 0 & \frac{\partial f(\gamma_2)\sin(\gamma_2)}{\partial \gamma_2} \\ 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}$$

3. Actuator term

In Figure 3 we illustrate a two dimensional schematic representation of a turbine with yaw angle γ and wind direction at the turbine rotor defined by the angle ϕ.

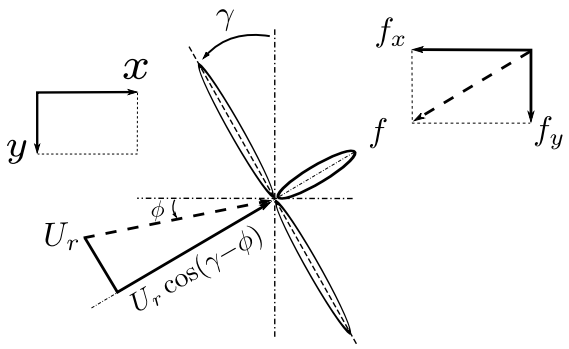


Figure 3: Schematic representation of a turbine with yaw angle γ, wind direction angle ϕ and rotor velocity U_r.

According to momentum theory, the following force term can be defined:

$$S_k = C_T \frac{1}{2} \rho A_r U_{\text{effect}}^2, \tag{13}$$

with thrust coefficient C_T, rotor swept area A_r, air density $\rho = 1.2$ and rotor upwind velocity U_{effect}. In actuator disk theory, the thrust coefficient is defined as $C_T = 4a(1-a)$ with a the turbines axial induction, γ its yaw angle and ϕ the wind direction at the rotor. However, the authors in [**?**] show that for high axial inductions, this thrust coefficient is not accurate with respect to measurements and a Glauert correction is introduced. The following adaptation of the C_T definition is proposed in [**?**]:

$$C_T = \begin{cases} 4aF(1-a), & \text{if } 0 \leq a \leq 0.4 \\ \left(8/9 + (4F - 40/9)a + (50/9 - 4F)a^2\right) & \text{if } 0.4 < a < 1 \end{cases} \tag{14}$$

In this paper, we set $F = 2$. This does not only result in a relative increase of C_T for high axial inductions, but also for low axial inductions. Since U_{effect} is not well defined in a wind farm, it is interesting to write the force in terms of the rotor velocity. We define the following relations:

$$\beta = \frac{a}{1-a}, \qquad U_{\text{effect}} = \frac{U_r \cos(\gamma - \phi)}{1-a}, \qquad U_r = \sqrt{u_r^2 + v_r^2}, \tag{15}$$

with U_r the flow velocity vector at the rotor with direction defined by the angle ϕ. Note that u_r and v_r are velocity components at the rotor in x- and y-direction respectively. Substituting these relations in Eq.(13) yields the following expression for the force S_k:

$$S_k = \frac{1}{2} \rho C_T A_r \left[U_r \cos(\gamma - \phi)(\beta + 1)\right]^2 \tag{16}$$

Now we can define the force in the x- and y-direction as

$$S_k^x = -S_k \cos(\gamma), \qquad S_k^y = S_k \sin(\gamma) \tag{17}$$

Power

The power production of a single turbine per cell is evaluated using the following expression:

$$P = 2\rho\pi A_r \beta \text{mean}(U)^3 \tag{18}$$